Lean-Backed NS Interface Note

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Abstract

This note records only the Navier–Stokes material currently backed by Lean in the MeTTapedia repository. It intentionally omits manuscript-specific claims about Ben Goertzel's December 11, 2025 SG-Cole-Hopf draft unless they have been encoded and checked in Lean.

The present Lean status is interface-level rather than proof-complete. The formalized material isolates reusable shells for cutoff continuity, topological Cole–Hopf transport, approximation interfaces, finite identity-energy algebra, abstract vorticity bounds, a kernel-semigroup wrapper, and two small concrete models showing both positive approximation instantiations and wrong topology obstructions, including one exact same-state-space split where a summable geometric observable is continuous and a stronger tail detector is not. It also now contains an abstract good observable class, an abstract tail-spike no-go theorem, and a first concrete soft-vs-hard chart fork on that same shared mode space, together with more manuscript-shaped Sobolev-style and energy-style soft-vs-hard forks on that same space, plus a log-energy fork showing that a smoother radial profile still lands on the same good/bad split, and finally a generic energy-profile theorem packaging that split for any continuous radial profile dominated by the identity on nonnegative inputs. It does *not* yet formalize the manuscript's specific chart, local hypotheses $(H1)$, $(H2)$, or the BKM-closing analytic bridge.

1 Scope

The Navier–Stokes Lean work currently lives in:

1. `FluidDynamics/NavierStokes/CutoffContinuity.lean`,
2. `FluidDynamics/NavierStokes/WeakTopologyCountermodel.lean`,
3. `FluidDynamics/NavierStokes/ProductTopologyL1Countermodel.lean`,
4. `FluidDynamics/NavierStokes/ColeHopfTopology.lean`,
5. `FluidDynamics/NavierStokes/ApproximationInterface.lean`,
6. `FluidDynamics/NavierStokes/ChartApproximationModel.lean`,
7. `FluidDynamics/NavierStokes/GeometricModeApproximation.lean`,
8. `FluidDynamics/NavierStokes/ModeStateProductTopologyObstruction.lean`,
9. `FluidDynamics/NavierStokes/ModeStateProductTopologyContinuity.lean`,
10. `FluidDynamics/NavierStokes/ModeStateObservableClass.lean`,
11. `FluidDynamics/NavierStokes/ModeStateTailSensitivityObstruction.lean`,

12. `FluidDynamics/NavierStokes/ModeStateChartFork.lean`,
13. `FluidDynamics/NavierStokes/ModeStateSobolevChartFork.lean`,
14. `FluidDynamics/NavierStokes/ModeStateEnergyChartFork.lean`,
15. `FluidDynamics/NavierStokes/ModeStateLogEnergyChartFork.lean`,
16. `FluidDynamics/NavierStokes/ModeStateEnergyProfileFork.lean`,
17. `FluidDynamics/NavierStokes/IdentityEnergy.lean`,
18. `FluidDynamics/NavierStokes/ColeHopfInterface.lean`,
19. `FluidDynamics/NavierStokes/KernelSemigroupInterface.lean`.

These modules compile without `sorry`. They provide a formal shell for what a future SG-Cole-Hopf proof attempt would need to instantiate.

2 Lean-Backed Results

2.1 Cutoff Continuity Shell

In `CutoffContinuity.lean`, the generic cutoff-shaped potential

$$\mathrm{cutoffPotential}(x) = \mathrm{cutoff}(\mathrm{radius}(x)) \cdot \mathrm{statistic}(x)$$

is formalized together with:

1. continuity of the cutoff potential when the scalar pieces are continuous;
2. a sequential obstruction theorem: if the cutoff factor and statistic converge to the wrong product, then the cutoff potential cannot converge to its claimed value;
3. the corresponding pointwise non-continuity criterion.

So Lean already captures the generic shape of a chart-cutoff continuity argument and its failure mode.

2.2 Concrete Weak-Topology Countermodel

In `WeakTopologyCountermodel.lean`, the abstract cutoff/Cole–Hopf shell is instantiated on the concrete state space $\mathbb{R} \times \mathbb{R}$ equipped with the topology induced by the first coordinate. Lean proves:

1. a sequence can converge in the ambient topology while changing a hidden coordinate;
2. a cutoff potential depending on that hidden coordinate is not continuous;
3. the associated Cole–Hopf transform is likewise not continuous.

This is a formal countermodel for the generic wrong-topology failure mode: ambient convergence can be too weak for the cutoff datum it is supposed to control.

2.3 Infinite-Mode Product-Topology Countermodel

In `ProductTopologyL1Countermodel.lean`, Lean moves the same mismatch to the raw infinite-mode sequence space $\mathbb{N} \to \mathbb{R}$ with the product topology. It proves:

1. finite-mode truncations converge coordinatewise in the product topology;
2. a tail spike moving to higher and higher coordinates converges to the zero sequence in that topology;
3. the stronger tail-sensitive observable $\sum_n |x_n|$ stays equal to 1 on that spike sequence and is therefore not continuous at zero;
4. the associated Cole–Hopf datum is likewise not continuous.

This is the first Lean-backed infinite-mode obstruction in the NS lane that directly separates weak coordinatewise convergence from a stronger tail norm.

2.4 Topological Cole–Hopf Shell

In `ColeHopfTopology.lean`, Lean formalizes:

1. continuity of the scalar map $y \mapsto \exp(-y/(2\nu))$;
2. continuity of the induced field-level Cole–Hopf transform;
3. injectivity when $\nu \neq 0$;
4. transfer of potential-limit mismatch into Cole–Hopf-limit mismatch;
5. the same obstruction specialized to cutoff-shaped potentials.

This means the topological $W \mapsto \phi$ step is already packaged in Lean, provided one supplies continuity or convergence of the underlying potential.

2.5 Approximation Interface

In `ApproximationInterface.lean`, Lean packages an abstract positive repair obligation:

> if an approximation scheme makes the chart radius observable and the chart statistic converge, then the cutoff potential and the Cole–Hopf datum also converge.

This is useful because it isolates exactly what an approximation/truncation theorem would have to prove in order for the cutoff/Cole–Hopf layer to behave well.

2.6 Concrete Chart/Truncation Model

In `ChartApproximationModel.lean`, the abstract approximation interface is instantiated on the concrete chart state space $\mathbb{R} \times \mathbb{R}$ with:

1. a hidden-coordinate radius observable;
2. a simple mixed chart statistic;
3. a truncation map that drops the hidden coordinate only at stage 0 and is exact thereafter.

Lean then proves:

1. convergence of the radius observable under truncation;
2. convergence of the chart statistic under truncation;
3. the resulting convergence of the cutoff potential;
4. the resulting convergence of the induced Cole–Hopf datum.

So the approximation interface is now backed by a concrete positive model, not just an abstract shell.

2.7 Geometric Finite-Mode Approximation Model

In `GeometricModeApproximation.lean`, the same approximation interface is instantiated on bounded infinite coefficient sequences. Lean formalizes:

1. a geometrically weighted mode radius;
2. a chart statistic built from the zeroth visible mode plus that radius;
3. finite-mode truncation keeping only the first N coefficients;
4. summability of the weighted mode series under the uniform coefficient bound;
5. convergence of the weighted radius under finite-mode truncation;
6. convergence of the derived chart statistic, the cutoff potential, and the induced Cole–Hopf datum.

This is the first genuinely infinite-mode positive instantiation in the NS Lean lane: the approximation is not just "wrong at stage 0, exact thereafter," but a real truncation-to-limit model on an infinite sequence state space.

2.8 Same-State-Space Product-Topology Obstruction

In `ModeStateProductTopologyObstruction.lean`, Lean returns to the exact `ModeState` and `truncateModes` objects from `GeometricModeApproximation.lean`, but equips that same state space with the coordinatewise product topology induced by the coefficient map. It proves:

1. the existing finite-mode truncations `truncateModes` do converge to the target state in this product topology;
2. a spike sequence still converges to the zero state in that same topology;
3. a stronger tail-presence observable, which records whether any nonzero coefficient exists, is not continuous there;
4. the associated Cole–Hopf datum is likewise not continuous.

So Lean now contains a positive and a negative result on one common infinite-mode object: the geometric weighted radius behaves well under truncation, while a stronger tail-sensitive observable on the same state space does not.

2.9 Same-State-Space Product-Topology Continuity

In `ModeStateProductTopologyContinuity.lean`, Lean proves the matching positive theorem on the exact same `ModeState` equipped with the exact same coordinatewise product topology:

1. each coordinate observable $x \mapsto x_n$ is continuous;
2. each weighted mode term $x \mapsto |x_n|2^{-n}$ is continuous;
3. the summable geometric radius $\sum_n |x_n|2^{-n}$ is continuous;
4. the derived chart statistic, cutoff potential, and Cole–Hopf datum are therefore continuous as well.

So the current NS Lean lane now cleanly separates two observables on one common infinite-mode state space: a summably dominated geometric radius that behaves well in the product topology, and a stronger tail detector that does not.

2.10 Abstract Good Observable Class

In `ModeStateObservableClass.lean`, Lean packages the good side on the shared `ModeState`: nonnegative summable mode weights. For that whole class, Lean proves:

1. continuity of the weighted radius observable;
2. continuity of the derived statistic, cutoff potential, and Cole–Hopf datum;
3. convergence of all of these along the existing truncations `truncateModes`;
4. recovery of the earlier geometric radius model as a special case.

So the good side is no longer just one geometric example. It is now an abstract summable observable class on the shared mode space.

2.11 Abstract Tail-Spike No-Go Theorem

In `ModeStateTailSensitivityObstruction.lean`, Lean packages the bad side abstractly too. If an observable is constant on every remote single-spike state `tailSpikeMode`(N) but takes a different value at `modeZero`, Lean proves:

1. the raw observable is not continuous at `modeZero`;
2. the same obstruction lifts to cutoff potentials;
3. the same obstruction lifts further to Cole–Hopf data.

So the bad side is no longer tied only to the one hand-built `tailPresenceRadius`. It is now a reusable theorem pattern.

2.12 First Concrete Chart Fork

In `ModeStateChartFork.lean`, Lean forces the first explicit chart-style fork on the shared mode space:

1. the soft support observable $t \mapsto |t|/(1+|t|)$, combined with summable mode weights, lies on the good side and yields a continuous truncation-friendly chart radius, statistic, cutoff potential, and Cole–Hopf datum;

2. the hard support observable, which only asks whether any coefficient is nonzero, lies on the bad side and fails continuity at `modeZero`, together with its cutoff and Cole–Hopf lifts.

   This is the first genuine theorem-level NS fork in Lean:

   > soft support belongs to the good summable side, hard support belongs to the bad tail-sensitive side.

2.13 Sobolev-Style Chart Fork

In `ModeStateSobolevChartFork.lean`, Lean pushes that fork one step closer to manuscript shape by amplifying each mode by the polynomial factor $((n+1)^m)$ before applying the chart cutoff:

1. the soft Sobolev observable, built from the bounded scalar $t \mapsto |t|/(1+|t|)$ together with a concrete inverse-square envelope, is continuous and truncation-friendly;

2. the corresponding hard Sobolev observable, which only asks whether any amplified mode is nonzero, is proved equal to the earlier hard support detector and is therefore not continuous at `modeZero`, together with its cutoff and Cole–Hopf lifts.

   So the current Lean fork is now sharper:

   > even after Sobolev-style high-frequency amplification, the soft bounded version lands on the good summable side, while the hard support version still lands on the bad side.

2.14 Energy-Style Chart Fork

In `ModeStateEnergyChartFork.lean`, Lean pushes the same shared-state fork one step further toward an energy-style observable:

1. the soft energy observable squares the Sobolev-amplified modes and then applies a compensating inverse-power envelope; Lean proves that this radius is continuous and truncation-friendly, and so are its cutoff and Cole–Hopf lifts;

2. the hard energy observable asks whether any amplified squared mode crosses the threshold 1; Lean proves that this detector is stable on the remote spike family, disagrees at `modeZero`, and is therefore not continuous, again together with its cutoff and Cole–Hopf lifts.

   So the current shared-mode-space fork is now sharper:

   > if the chart observable keeps enough inverse-power compensation to remain summable after Sobolev amplification and squaring, Lean puts it on the good approximation-friendly side; if it turns into a hard energy threshold on those amplified modes, Lean puts it on the bad tail-sensitive side.

2.15 Log-Energy Chart Fork

In `ModeStateLogEnergyChartFork.lean`, Lean tests a smoother radial profile on the same shared mode space by replacing raw squared Sobolev energy by $\log(1 + \text{energy})$:

1. the soft log-energy observable is continuous and truncation-friendly; the key formal domination is $\log(1 + u) \leq u$, which lets Lean compare the log-energy terms to the already-controlled soft energy terms;

2. the hard-side analogue asking whether $\log(1 + \text{energy}) > 0$ still lands on the bad side: it is stable on the remote spike family, disagrees at `modeZero`, and therefore its cutoff and Cole–Hopf lifts are also not continuous.

   So the shared-mode-space fork is now sharper again:

   > even replacing raw energy by the smoother profile $\log(1 + \text{energy})$ does not escape the same dichotomy: the summably compensated soft version stays on the good side, while the hard detection version stays on the bad side.

2.16 Generic Energy-Profile Fork

In `ModeStateEnergyProfileFork.lean`, Lean abstracts the preceding examples into one theorem pattern. An `EnergyProfile` is a globally continuous radial profile $\rho : \mathbb{R} \to \mathbb{R}$ satisfying, on nonnegative inputs:

1. $0 \leq \rho(u)$,

2. $\rho(u) \leq u$,

3. $\rho(0) = 0$,

4. $u > 0 \Rightarrow \rho(u) > 0$.

   For every such profile, Lean proves:

1. the summably compensated soft radius is continuous and truncation-friendly;

2. the induced hard detector agrees with the raw hard Sobolev support detector;

3. that hard detector is tail-spike stable and therefore not continuous at `modeZero`, together with its cutoff and Cole–Hopf lifts.

   The file also includes canonical instances:

1. the identity profile, recovering the raw soft energy radius;

2. the clipped log profile $u \mapsto \log(1 + \max(u, 0))$, recovering the soft and hard log-energy constructions on the nonnegative energy inputs that actually occur;

3. a bounded rational saturation profile $u \mapsto \max(u, 0)/(1 + \max(u, 0))$, which is closer to a chart-cutoff saturation shape and whose hard detector Lean proves collapses back to the hard Sobolev support detector;

4. a smooth exponential saturation profile $u \mapsto 1 - \exp(-\max(u, 0))$, again with the same hard-detector collapse;

5. a smooth-transition profile $u \mapsto u \cdot \mathrm{smoothTransition}(u)$, using mathlib's `Real.smoothTransition`, again with the same hard-detector collapse;

6. more strongly, a positive-parameter family $u \mapsto u \cdot \mathrm{smoothTransition}(cu)$ for $c > 0$, all with the same hard-detector collapse.

So the current shared-mode-space NS fork is now theorem-level rather than example-level:

> every continuous radial profile that stays nonnegative, vanishes only at zero, and is dominated by the identity on nonnegative inputs lands on the same good/bad fork.

2.17 Finite Identity-Energy Algebra

In `IdentityEnergy.lean`, Lean formalizes a finite carré-du-champ-style energy

$$\gamma(D) = \sum_i D_i^2$$

and proves:

1. basic positivity and scaling laws for γ;
2. the log-gradient comparison used in the identity-only Cole–Hopf calculation;
3. the finite Cauchy–Schwarz frame-evaluation bound;
4. the resulting pointwise vorticity-style estimate after push-down through a curl frame.

So the finite algebraic spine of the identity-only bound is already Lean-backed.

2.18 Abstract Cole–Hopf/Vorticity Interface

In `ColeHopfInterface.lean`, the finite identity-energy algebra is packaged into an abstract structure carrying:

1. uniform positivity of $\Phi(t)$;
2. a uniform identity-energy bound on its directional derivatives;
3. a uniform frame-curl bound.

From those hypotheses, Lean derives:

1. a bound on the log-potential coefficients $W(t) = -2\nu \log \Phi(t)$;
2. a pointwise abstract vorticity bound;
3. the uniform-in-time version of the same estimate.

2.19 Kernel-Semigroup Wrapper

In `KernelSemigroupInterface.lean`, the same identity-only Cole–Hopf/vorticity algebra is wrapped around a local Markov-kernel semigroup. This does not add new analytic content. It says that if a future NS evolution is encoded as a local kernel semigroup, then the existing identity-energy bridge can be reused unchanged.

3 Current Boundary

The following are *not* yet formalized in Lean in the present NS lane:

1. the manuscript's specific chart-cutoff datum on SDiff^m;
2. the finite-mode truncation / approximation theorem actually needed for that datum;
3. the local hypotheses $(H1)$, $(H2)$;
4. the manuscript's claimed identity-only propagation lemma;
5. the BKM-closing bridge for the full infinite-dimensional argument.

So the honest current status is:

> Lean has formalized the generic interface shell that a proof attempt would need, but not the manuscript-specific analytic instantiations.

4 Verification

The currently relevant NS modules were checked with:

```
lake build \
  Mettapedia.FluidDynamics.NavierStokes.ModeStateEnergyProfileFork \
  Mettapedia.FluidDynamics.NavierStokes.ModeStateLogEnergyChartFork \
  Mettapedia.FluidDynamics.NavierStokes.ModeStateEnergyChartFork \
  Mettapedia.FluidDynamics.NavierStokes.ModeStateSobolevChartFork \
  Mettapedia.FluidDynamics.NavierStokes.ModeStateChartFork \
  Mettapedia.FluidDynamics.NavierStokes.ModeStateTailSensitivityObstruction \
  Mettapedia.FluidDynamics.NavierStokes.ModeStateObservableClass \
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  Mettapedia.FluidDynamics.NavierStokes.ChartApproximationModel \
  Mettapedia.FluidDynamics.NavierStokes.IdentityEnergy \
  Mettapedia.FluidDynamics.NavierStokes.WeakTopologyCountermodel \
  Mettapedia.FluidDynamics.NavierStokes.KernelSemigroupInterface \
  Mettapedia.FluidDynamics.NavierStokes.ApproximationInterface \
  Mettapedia.FluidDynamics.NavierStokes.CutoffContinuity \
  Mettapedia.FluidDynamics.NavierStokes.ColeHopfTopology \
  Mettapedia.FluidDynamics.NavierStokes.ColeHopfInterface
```

5 Next Lean Target

The natural next formal step is no longer more prose. It is to instantiate one of the current abstract fork with manuscript-specific data:

1. formalize a manuscript-shaped chart observable on the shared mode space whose shape is *closer* than the current Sobolev-style soft/hard toy, and prove whether it belongs to the good summable class or the bad tail-sensitive class;

2. if it lands on the good side, formalize the approximation/truncation theorem needed to feed `ApproximationInterface.lean`;
3. formalize the manuscript's local identity-energy bridge inside the abstract shell already provided by `IdentityEnergy.lean` and `ColeHopfInterface.lean`.